

Restoration and Sketch Studio: Universal, Hard-Routed, and Soft Mixture-of-Experts Image Restoration with a Conditional Face-to-Sketch GAN

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Generative AI Assignment 1, Section A

Abstract—This report presents four generative models served through one containerized web application. Three image restoration systems are trained on the Oxford-IIIT Pet dataset to remove salt-and-pepper noise, Gaussian blur, and rectangular occlusion. A universal autoencoder handles every condition with one shared bottleneck. A hard-routed system uses a classifier to select one of three specialist autoencoders. A soft mixture of experts learns the routing jointly. A fourth model is a style-conditioned conditional GAN that turns face photos into sketches in three styles using the FS2K dataset. Every model is tuned with Optuna, tracked with MLflow, exported to ONNX, and served by a FastAPI backend behind a React interface. On 36,690 fixed test entries the soft mixture reaches 21.54 dB PSNR and 0.692 SSIM and is the only restoration system that beats returning the corrupted input (19.70 dB, 0.639). The universal and hard-routed autoencoders score below that baseline, and the report analyses why. The classifier reaches 0.998 accuracy. The sketch generator reaches 0.459 SSIM on 1,045 official test pairs.

Index Terms—denoising autoencoder, mixture of experts, conditional GAN, Optuna, ONNX, image restoration, face sketch synthesis

I. INTRODUCTION

Image restoration models are usually trained for one degradation. Real inputs are mixed, so a deployed system must either handle all degradations with one network, decide which specialist to call, or blend specialists. This work builds all three strategies on the same data, the same fixed test protocol, and the same metrics, and adds a fourth generative task, face-to-sketch synthesis, that uses an adversarial objective. The goal is not only to report scores but to explain what each design choice did to the result.

The main contributions are as follows.

- A reproducible corruption pipeline with fixed validation and test manifests, so every model is compared on identical corrupted images.
- A universal denoising autoencoder with a true vector bottleneck and no skip connections (Task 1).
- A classifier plus three specialists with oracle and predicted routing, and clean images bypassing restoration (Task 2).
- A soft mixture of experts whose gate is initialised from the Task 2 classifier and fine-tuned jointly, with a routing analysis (Task 3).
- A style-conditioned pix2pix-style GAN for FS2K sketches (Task 4).

- An Optuna study per task, MLflow tracking, ONNX export, and a Docker Compose application with four workspaces.

The code is at <https://github.com/AbdullahRasheed452/GENAI-1>. The demonstration video is at <https://youtu.be/kPPI9NkhYq4>.

II. RELATED WORK

Denoising autoencoders learn to reconstruct clean data from corrupted input by passing it through a compressed code [1]. U-Net adds skip connections between matching encoder and decoder stages [2]. We deliberately avoid skips in Tasks 1 to 3 because the assignment requires a genuine bottleneck, and we use them in the Task 4 generator, where spatial detail transfer is desirable.

Mixtures of experts combine specialist networks through a learned gate [3]. Sparse variants route each input to a few experts at scale [4]. Our gate is dense and small, and it includes an identity branch that lets the model leave an input untouched.

Conditional GANs condition both generator and discriminator on side information [5], [6]. Pix2pix pairs a U-Net generator with a PatchGAN discriminator and an L1 term for paired translation [7]. We follow that design and add a learned style embedding. SSIM [8] is used both in the restoration loss and as an evaluation metric. Hyperparameters are searched with Optuna [9], and runs are logged with MLflow [10]. The datasets are Oxford-IIIT Pet [11] and FS2K [12].

III. DATA AND CORRUPTION PROTOCOL

A. Oxford-IIIT Pet

All 7,349 images are resized to 128×128 . The official test set of 3,669 images is kept for testing. The remaining 3,680 images are split with seed 42 into 2,944 for training and 736 for validation. No test image is used for any decision.

B. Corruptions

Corruptions are generated at run time during training with parameters drawn from the ranges in Table I. Salt-and-pepper noise flips a pixel to white or black with equal probability. Gaussian blur uses a separable kernel with reflection padding. Occlusion paints one to three black rectangles whose union covers a target fraction of the image, with aspect ratios between 0.5 and 2.0.

TABLE I
TRAINING CORRUPTION RANGES.

Corruption	Sampled range
Salt-and-pepper	probability 0.02 to 0.15
Gaussian blur	kernel 3, 5, or 7, sigma 0.5 to 2.5
Occlusion	coverage 0.10 to 0.35, 1 to 3 rectangles

TABLE II
FIXED TEST SEVERITIES.

Corruption	Low	Medium	High
Salt-and-pepper	$p=0.03$	$p=0.08$	$p=0.15$
Gaussian blur	$k=3, \sigma=0.7$	$k=5, \sigma=1.5$	$k=7, \sigma=2.5$
Occlusion	10%, 1 box	20%, 2 boxes	35%, 3 boxes

Validation and test use fixed manifests. The validation manifest has 736 entries. The test manifest has 36,690 entries, which is 3,669 images times ten conditions: one clean version and three severities of each of the three corruptions (Table II). The manifests are stored in the repository, so any result can be reproduced exactly.

C. FS2K

FS2K provides face photos with sketches in three styles. The official train and test split is taken from its annotation files. The test set has 1,045 pairs because one file is missing from the downloaded copy, with 618 Style 1, 381 Style 2, and 46 Style 3 pairs. Style 3 is therefore rare, and its scores are noisy. Images are resized to 128×128 .

D. Metrics

Restoration is scored with PSNR in dB and SSIM against the clean image. PSNR is capped at 100 dB so that exact reconstructions do not produce infinite values. A reference row named Input scores the corrupted image itself, which is what a model that does nothing would return. Any restoration system must beat this row to be useful. Sketches are scored with L1, SSIM, and PSNR against the ground truth sketch.

IV. SYSTEM OVERVIEW

Figure 1 summarises the four models. All models are trained in PyTorch on free notebook GPUs, exported to ONNX, and served on CPU through ONNX Runtime. Every Optuna study uses the TPE sampler with a median pruner, and every trial and final run is logged to MLflow in SQLite databases that are stored in the repository.

V. TASK 1: UNIVERSAL DENOISING AUTOENCODER

A. Architecture and loss

The encoder has four stages. Each stage is a stride-2 convolution block followed by a second convolution block, with batch normalisation and ReLU. Channels double at each stage from the base width. The final 8×8 feature map is flattened and projected by a linear layer to a latent vector, followed by dropout. The decoder mirrors this with a linear layer, four stages of nearest-neighbour upsampling and convolution, and

a sigmoid output. With a latent size of 256, a $128 \times 128 \times 3$ image of 49,152 values is compressed 192 times. There are no skip connections, so all information must pass through the vector.

The loss combines pixel and structural terms,

$$\mathcal{L} = \alpha \|x - \hat{x}\|_1 + (1 - \alpha) (1 - \text{SSIM}(x, \hat{x})), \quad (1)$$

where x is the clean target and $\hat{x} = D(E(\tilde{x}))$ is the reconstruction of the corrupted input $\tilde{x}$. Training uses Adam with cosine annealing. The model is never told which corruption was applied. Training draws clean, noisy, blurred, and occluded images at random.

B. Optuna search

The search space covers learning rate (log scale), batch size (16, 32, 64), latent dimension (256, 512, 1024), base channels (32, 48, 64), dropout, and α . The validation objective is $0.5 \text{L1} + 0.5 (1 - \text{SSIM})$ on the validation manifest. The study ran 20 trials of 10 epochs, with 14 completed and 6 pruned. The best trial was number 15 with objective 0.3102. The selected configuration is learning rate 2.63×10^{-4} , batch size 16, latent 256, base 48, dropout 0.032, and $\alpha = 0.508$ (Table XIII). The search range for α was 0.5 to 0.95. Smaller latents did well, and the best completed trials used latent 256 with α between 0.50 and 0.63. The suggested starting value of 0.8 was not selected, which shows that the SSIM term matters. Because the best values sit at the lower edge of the range, a lower α might do even better, and this was not tested. Figure 3 shows the optimisation history.

C. Training and results

The selected configuration was trained twice with identical settings, once for 60 epochs and once for 30 epochs, and both runs are recorded in MLflow. The 30 epoch run reached its best validation objective of 0.2985 at epoch 18, and the 60 epoch run reached 0.2981 at epoch 16. In both runs the validation objective stops improving at about that point while the training loss keeps falling. In the 60 epoch run the training loss falls from 0.2688 at epoch 16 to 0.1849 at epoch 60 while the validation objective rises to 0.3113, which is overfitting. The checkpoint with the best validation objective is therefore the one used. Figure 2 shows the curves of the 30 epoch run.

Table III reports the fixed test results. The most important finding is that the model scores between 17.3 and 18.9 dB on every condition, almost independent of the corruption and its severity. Salt-and-pepper stays at 18.80 to 18.84 dB across all three levels. This is the reconstruction ceiling of the bottleneck, because even clean images come back at 18.82 dB. The model therefore helps only when the corrupted input is worse than that ceiling. It improves salt-and-pepper at medium and high severity (15.89 to 18.84 dB and 13.16 to 18.80 dB) and occlusion at every severity (16.57 to 18.43 dB at low and 10.75 to 17.31 dB at high). It loses to the do-nothing input on blur (29.14 to 18.89 dB), on low-noise salt-and-pepper (20.15 to 18.84 dB), and on clean images. Averaged over all corrupted entries it reaches 18.55 dB and 0.487 SSIM against 19.70 dB and 0.639 for the input.

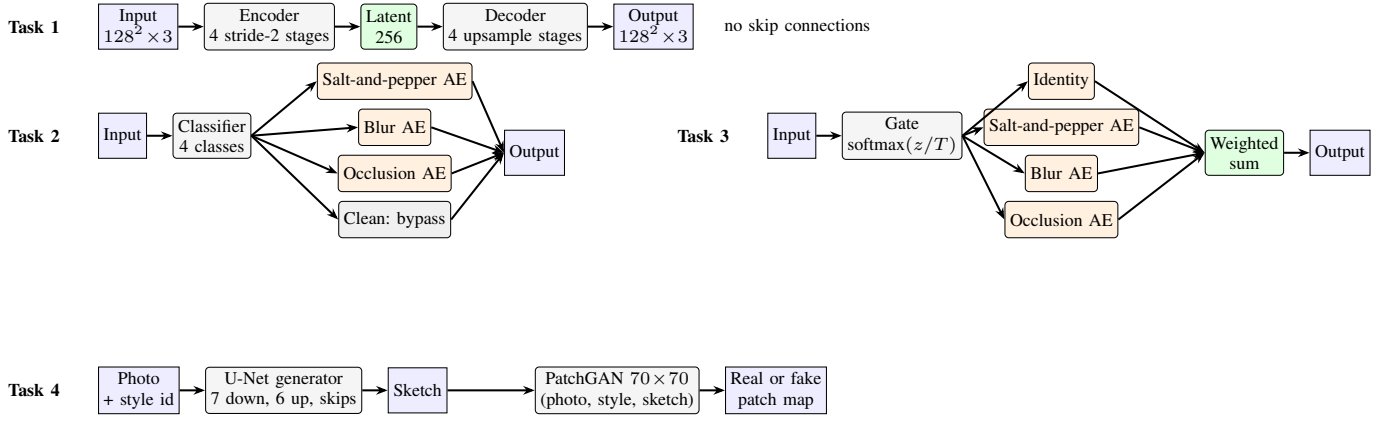


Fig. 1. Architectures of the four tasks. Task 3 reuses the Task 2 classifier as the gate and the three specialists as experts.

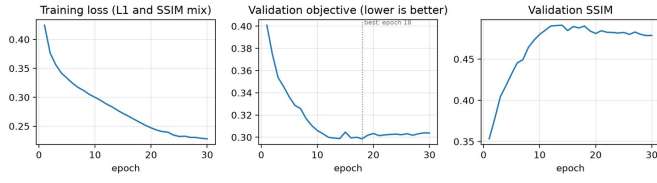


Fig. 2. Task 1 training and validation curves of the 30 epoch run. Validation stops improving near epoch 18 while training loss keeps falling.

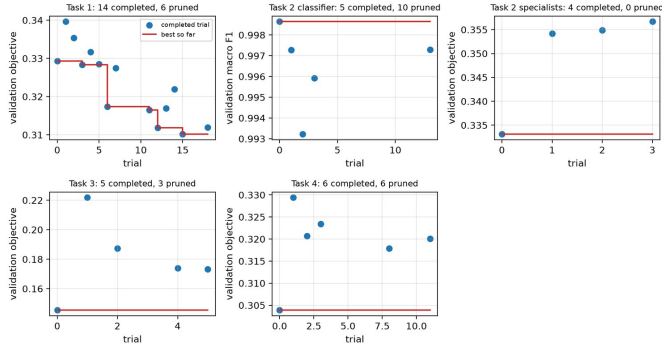


Fig. 3. Optuna optimisation history for the studies.

D. Examples and failures

Figure 4 shows examples with the clean target, corrupted input, reconstruction, and an absolute error map, and Fig. 5 shows four failure cases. The reconstructions are smooth and lose colour saturation, drifting toward a brown tone. The error maps show that the error concentrates on texture such as stone, foliage, and fur, which a 256-dimensional code cannot store. In the low occlusion failure the clean target is mostly black background with a pale cat, and the model does not recover the sharp silhouette. In the high occlusion failure the hidden region is filled with blurred average colour instead of plausible content. These failures follow directly from the bottleneck and the L1 and SSIM objective, which favour blurry averages when information is missing.

TABLE III
TASK 1 TEST RESULTS BY CONDITION AND SEVERITY (PSNR IN DB, SSIM). IN. IS THE SCORE OF THE CORRUPTED INPUT.

Condition	Severity	PSNR	SSIM	In. PSNR	In. SSIM
Clean	—	18.82	0.497	—	1.000
Salt-and-pepper	Low	18.84	0.497	20.15	0.588
Salt-and-pepper	Medium	18.84	0.495	15.89	0.322
Salt-and-pepper	High	18.80	0.492	13.16	0.190
Salt-and-pepper	All	18.83	0.495	16.40	0.367
Gaussian blur	Low	18.85	0.498	34.06	0.958
Gaussian blur	Medium	18.89	0.497	28.06	0.841
Gaussian blur	High	18.94	0.495	25.32	0.731
Gaussian blur	All	18.89	0.496	29.14	0.843
Occlusion	Low	18.43	0.486	16.57	0.859
Occlusion	Medium	18.01	0.473	13.37	0.726
Occlusion	High	17.31	0.453	10.75	0.539
Occlusion	All	17.92	0.471	13.56	0.708
All corrupted	All	18.55	0.487	19.70	0.639

VI. TASK 2: HARD-ROUTED RESTORATION

A. Design

A small convolutional classifier predicts one of four classes: clean, salt-and-pepper, blur, or occlusion. It uses four double-convolution blocks with max pooling, global average pooling, dropout, and a linear head. It is trained with cross entropy using a balanced batch sampler so that every class appears equally. Three specialist autoencoders with the Task 1 architecture are each trained on one corruption only. A clean prediction bypasses restoration and returns the input unchanged. Routing is evaluated in two ways. Oracle routing uses the true label, and predicted routing uses the classifier.

B. Optuna and training

The classifier search tried learning rate, batch size, channel width (small, medium, large), dropout, and weight decay over 15 trials, with 5 completed and 10 pruned. The best trial had a validation macro F1 of 0.9986 and chose the small network with batch size 16 and dropout 0.433. The specialist study shared one configuration across the three experts. It started 5

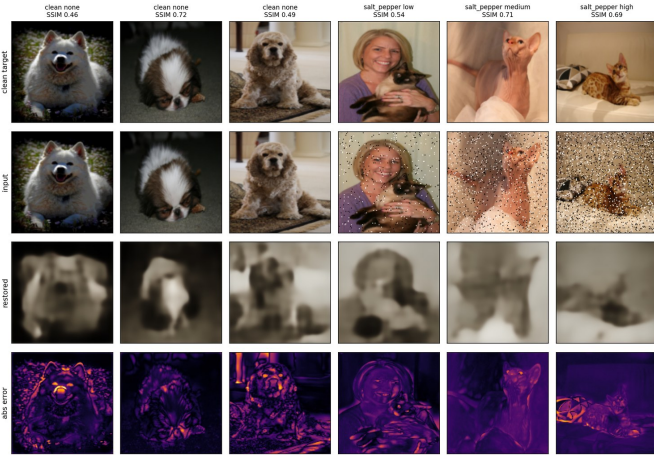


Fig. 4. Task 1 examples: clean target, corrupted input, restored output, and absolute error map.

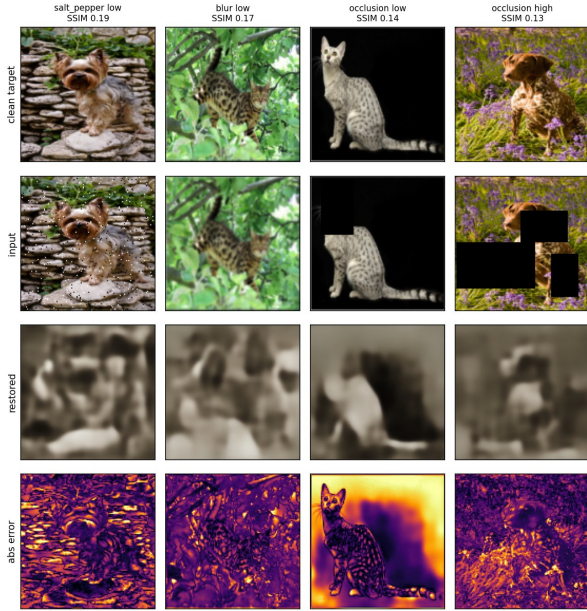


Fig. 5. Task 1 failure cases. SSIM is shown above each column.

trials, of which 4 completed and 1 failed when the notebook session was reset, so this study is smaller than planned. The best of the four completed trials had objective 0.3331 and used latent 256, base 32, and $\alpha = 0.509$. The final classifier and the three specialists were trained for 30 and 40 epochs respectively (Figs. 6 and 7).

C. Classification results

On 36,690 test entries the classifier reaches accuracy 0.9980 and macro F1 0.9967 (Table IV, Fig. 8). Salt-and-pepper is perfectly classified. The weakest case is low severity occlusion, with recall 0.9869. In total 48 occluded images were called clean, 15 clean images were called blur, and 8 clean images were called occlusion. These are the cases where a small hidden region barely changes the image.

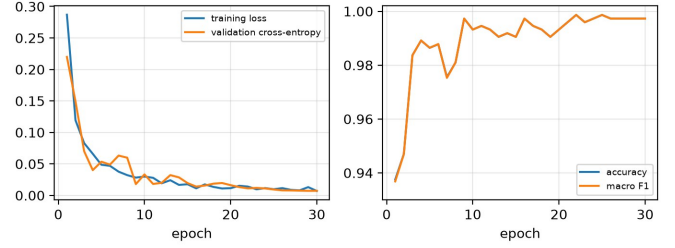


Fig. 6. Classifier training curves.

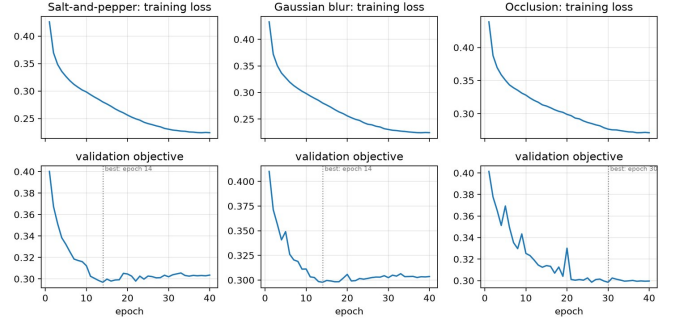


Fig. 7. Specialist training curves.

D. Restoration results

Table V gives the predicted-routing results and Table VI gives the oracle results. The two are almost identical (18.44 dB and 0.494 against 18.43 dB and 0.494), so classifier errors are not what limits the system. Perfect routing would not help. The experts themselves are the limit. Each specialist has the same bottleneck ceiling as Task 1, so specialisation barely helps: the average is 18.44 dB and 0.494 SSIM against 18.55 dB and 0.487 for the universal model, which is a negligible difference. The practical gain is on clean images, which the bypass returns at 99.49 dB, where the universal model returns 18.82 dB. The system still scores below the corrupted-input baseline on average and loses badly on blur (18.61 against 29.14 dB), because it replaces a mild degradation with a worse reconstruction.

Figure 9 shows examples. Figure 10 shows the four worst failures, and all four are clean images that were routed to a specialist by mistake and then came back blurred, with SSIM between 0.18 and 0.29. Each of these images has large black regions, either a dark background or black bars at the top and bottom. Black areas resemble an occlusion box, which is the most likely reason for the confusion, and it matches the 8 clean images called occlusion and the 15 called blur. One wrong routing decision costs a lot, because the specialist destroys an image that was already perfect.

VII. TASK 3: SOFT MIXTURE OF EXPERTS

A. Design

The model has four branches: an identity branch and the three specialist autoencoders. A gate network with the

TABLE IV
CLASSIFIER TEST RESULTS.

Class	Precision	Recall	F1
Clean	0.9865	0.9937	0.9901
Salt-and-pepper	1.0000	1.0000	1.0000
Gaussian blur	0.9986	0.9998	0.9992
Occlusion	0.9993	0.9956	0.9975
Macro average	0.9961	0.9973	0.9967
Accuracy			0.9980

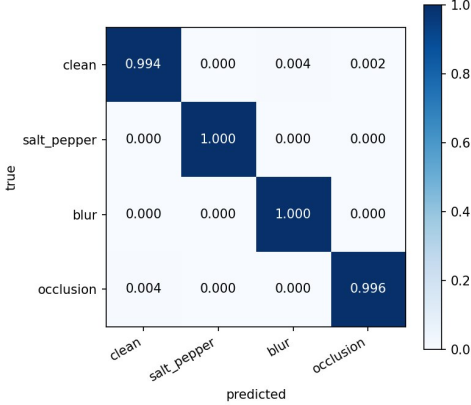


Fig. 8. Row-normalised confusion matrix of the corruption classifier.

classifier architecture produces logits z , and the weights are $w = \text{softmax}(z/T)$. The output is $\hat{x} = \sum_i w_i f_i(\tilde{x})$ with f_0 the identity. The gate starts from the trained Task 2 classifier and the experts start from the trained specialists. Training has two phases. In a warm-up of 2 epochs the experts are frozen and kept in evaluation mode while the gate adapts. In 15 epochs of joint fine-tuning all parts train together with

$$\mathcal{L} = \alpha \|x - \hat{x}\|_1 + (1 - \alpha)(1 - \text{SSIM}) + \lambda_{ce} \text{CE}(z/T, y) + \lambda_{bal} \sum_i (\bar{w}_i - 0.25)^2, \quad (2)$$

where $\bar{w}_i$ is the batch mean weight of branch i . The cross entropy term keeps the gate tied to the true corruption, and the balance term discourages a single branch from taking all the weight.

B. Optuna and training

The study searched learning rate, temperature T , λ_{ce} , λ_{bal} , and α over 8 trials, with 5 completed and 3 pruned. A trial is also pruned when one branch takes more than 0.7 of the mean weight, which is treated as routing collapse. The best trial had objective 0.1454 with learning rate 3.57×10^{-5} , $T = 0.5$, $\lambda_{ce} = 0.0205$, $\lambda_{bal} = 0.00205$, and $\alpha = 0.526$, again close to the lower edge of the 0.5 to 0.95 range. Objective values are not comparable across tasks, because the mixture can copy the input through the identity branch. Figure 11 shows the training curves.

TABLE V
TASK 2 TEST RESULTS WITH PREDICTED ROUTING.

Condition	Severity	PSNR	SSIM	In. PSNR	In. SSIM
Clean	–	99.49	0.997	–	1.000
Salt-and-pepper	Low	18.60	0.502	20.15	0.588
Salt-and-pepper	Medium	18.60	0.502	15.89	0.322
Salt-and-pepper	High	18.55	0.500	13.16	0.190
Salt-and-pepper	All	18.58	0.501	16.40	0.367
Gaussian blur	Low	18.52	0.503	34.06	0.958
Gaussian blur	Medium	18.61	0.502	28.06	0.841
Gaussian blur	High	18.69	0.499	25.32	0.731
Gaussian blur	All	18.61	0.501	29.14	0.843
Occlusion	Low	18.59	0.497	16.57	0.859
Occlusion	Medium	18.19	0.481	13.37	0.726
Occlusion	High	17.61	0.463	10.75	0.539
Occlusion	All	18.13	0.480	13.56	0.708
All corrupted	All	18.44	0.494	19.70	0.639

TABLE VI
TASK 2 TEST RESULTS WITH ORACLE ROUTING. THE ASTERISK MARKS THE 100 dB CAP ON EXACT RECONSTRUCTION.

Condition	Severity	PSNR	SSIM	In. PSNR	In. SSIM
Clean	–	100*	1.000	–	1.000
Salt-and-pepper	Low	18.60	0.502	20.15	0.588
Salt-and-pepper	Medium	18.60	0.502	15.89	0.322
Salt-and-pepper	High	18.55	0.500	13.16	0.190
Salt-and-pepper	All	18.58	0.501	16.40	0.367
Gaussian blur	Low	18.52	0.502	34.06	0.958
Gaussian blur	Medium	18.61	0.502	28.06	0.841
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Gaussian blur	All	18.61	0.501	29.14	0.843
Occlusion	Low	18.53	0.492	16.57	0.859
Occlusion	Medium	18.19	0.481	13.37	0.726
Occlusion	High	17.61	0.463	10.75	0.539
Occlusion	All	18.11	0.479	13.56	0.708
All corrupted	All	18.43	0.494	19.70	0.639

C. Results

Table VII reports the results. The mixture reaches 21.54 dB and 0.692 SSIM averaged over corrupted entries, against 19.70 dB and 0.639 for the input. It is the only system in this work that improves on the do-nothing baseline on average. The gains are large for salt-and-pepper (16.40 to 19.63 dB, SSIM 0.367 to 0.540) and occlusion (13.56 to 17.88 dB). On blur it still loses (29.14 to 27.10 dB, SSIM 0.843 to 0.815), but much less than Tasks 1 and 2. Clean images are returned at 51.31 dB.

D. Routing analysis

Table VIII and Fig. 12 show the mean weights per condition. The gate behaves sensibly. Clean images take 0.974 identity weight. Medium and high salt-and-pepper take about 1.0 on the salt-and-pepper expert. Occlusion weight grows with severity, from 0.139 to 0.393 to 0.711, and identity weight falls accordingly.

Two behaviours deserve attention. First, low severity salt-and-pepper sends 0.307 of its weight to identity. At that level

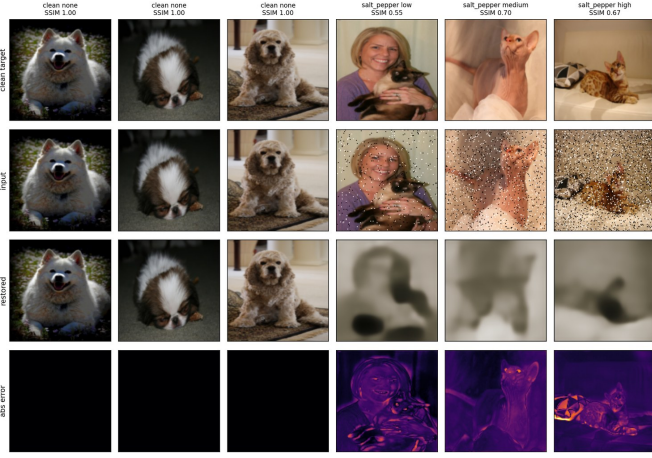


Fig. 9. Task 2 examples with predicted routing.

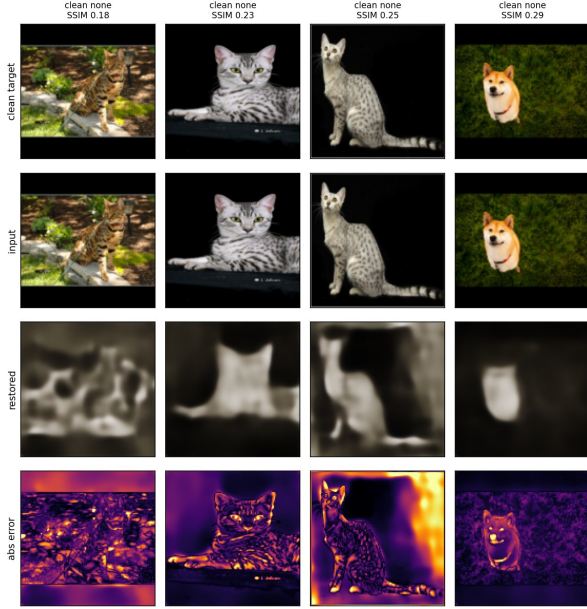


Fig. 10. Task 2 failure cases.

the input already scores 20.15 dB, which is better than the expert ceiling of about 18.6 dB, so keeping part of the input is the correct decision. Second, blur is routed mostly to identity (0.796 to 0.856) and the blur expert has a mean weight of only 0.055 and is never the top choice (Table IX). The gate has learned that blurred inputs score about 29 dB, so any expert that outputs 18.6 dB makes the result worse. This is a correct local decision, and it means the blur expert is effectively inactive. It also explains why the mixture beats the hard-routed system, which was forced to use that expert. The weight histograms (Fig. 13) show confident, near-binary weights for salt-and-pepper and spread weights for occlusion, and Fig. 14 shows cases where the gate splits its weight between branches.

Figures 15 and 16 show examples and failures. The failures fall into three kinds. A lightly corrupted dog image sent mostly

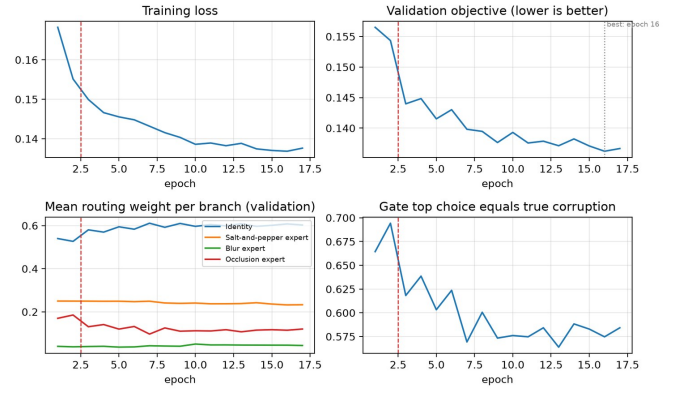


Fig. 11. Task 3 training curves with the warm-up and joint phases.

TABLE VII
TASK 3 TEST RESULTS.

Condition	Severity	PSNR	SSIM	In. PSNR	In. SSIM
Clean	—	51.31	0.999	—	1.000
Salt-and-pepper	Low	21.11	0.613	20.15	0.588
Salt-and-pepper	Medium	18.90	0.505	15.89	0.322
Salt-and-pepper	High	18.87	0.503	13.16	0.190
Salt-and-pepper	All	19.63	0.540	16.40	0.367
Gaussian blur	Low	30.61	0.935	34.06	0.958
Gaussian blur	Medium	26.30	0.806	28.06	0.841
Gaussian blur	High	24.38	0.705	25.32	0.731
Gaussian blur	All	27.10	0.815	29.14	0.843
Occlusion	Low	18.04	0.864	16.57	0.859
Occlusion	Medium	17.80	0.735	13.37	0.726
Occlusion	High	17.81	0.563	10.75	0.539
Occlusion	All	17.88	0.721	13.56	0.708
All corrupted	All	21.54	0.692	19.70	0.639

to the salt-and-pepper expert comes back as a blurry blob (SSIM 0.28), although the input was nearly intact, which is the cost of imperfect gating at low severity. A blurred cat image is returned almost as it came (SSIM 0.71) because the gate chose identity, so nothing was deblurred. In the medium occlusion case the identity branch keeps part of the black boxes visible while the expert fills only some of the hidden area, and under high noise the expert output collapses to soft colour (SSIM 0.16).

E. Comparison of the three strategies

Table X compares all three systems. The universal and hard-routed models behave almost the same, with 18.55 and 18.44 dB, because both are limited by the autoencoder bottleneck. The soft mixture wins by adding an identity path and by learning when not to restore. The comparison is sound because all systems use the same manifests. It has limits. Each system comes from one training run with one seed, so small differences, such as 0.11 dB between Tasks 1 and 2, should not be read as real.

TABLE VIII
MEAN ROUTING WEIGHTS PER CONDITION AND SEVERITY.

True condition	Severity	Identity	Salt-and-pepper	Blur	Occlusion
Clean	–	0.974	0.000	0.003	0.023
Salt-and-pepper	Low	0.307	0.607	0.008	0.078
Salt-and-pepper	Medium	0.002	0.995	0.001	0.002
Salt-and-pepper	High	0.000	1.000	0.000	0.000
Gaussian blur	Low	0.856	0.000	0.140	0.004
Gaussian blur	Medium	0.804	0.000	0.193	0.003
Gaussian blur	High	0.796	0.000	0.201	0.003
Occlusion	Low	0.858	0.000	0.003	0.139
Occlusion	Medium	0.604	0.001	0.002	0.393
Occlusion	High	0.285	0.002	0.002	0.711

TABLE IX
BRANCH HEALTH ON THE TEST SET.

Branch	Mean weight	Top choice	Weight on other types	Status
Identity	0.549	0.621	0.501	dominates other types
Salt-and-pepper	0.261	0.265	0.001	healthy
Gaussian blur	0.055	0.000	0.003	inactive
Occlusion	0.136	0.114	0.016	healthy

VIII. TASK 4: FACE-TO-SKETCH CONDITIONAL GAN

A. Architecture and loss

The generator is a U-Net with seven stride-2 down blocks and six up blocks with skip connections, taking a 128×128 photo and producing a tanh sketch. A learned embedding of the style id (1, 2, or 3) is concatenated to the input as extra channels and again at the bottleneck. Dropout is applied in the first three decoder blocks. The discriminator is a 70×70 PatchGAN that sees the photo, the style, and a real or generated sketch. The objective is

$$\mathcal{L}_G = \text{BCE}(D(x, s, G(x, s)), 1) + \lambda_{L1} \|y - G(x, s)\|_1, \quad (3)$$

and the discriminator minimises the usual BCE on real and fake pairs. Both use Adam with $\beta = (0.5, 0.999)$.

B. Optuna and training

The study ran 12 trials of 8 epochs, with 6 completed and 6 pruned, searching both learning rates, batch size, base width, dropout, embedding size, and λ_{L1} . The best trial (number 0, objective 0.3039) used $\text{lr}_G = 1.54 \times 10^{-4}$, $\text{lr}_D = 8.63 \times 10^{-4}$, batch size 8, base 64, dropout 0.301, embedding size 32, and $\lambda_{L1} = 121$. In one pruned trial the discriminator losses fell to about 0.007 within three epochs while the generator adversarial loss rose above 5, which is the typical failure where the discriminator wins. The final run trained for 60 epochs. The discriminator losses stayed near 0.4 to 0.5 for real and fake pairs, so training remained balanced (Fig. 17). The best validation objective was 0.3064 at epoch 57. This is slightly worse than the 8-epoch trial value of 0.3039, so longer training did not improve the pixel and SSIM objective. Figure 18 shows the sample sketches improving visually across epochs.

C. Results

Table XI gives the test results. Overall the generator reaches SSIM 0.459 and PSNR 15.48 dB. Style 1 reaches 0.500 and

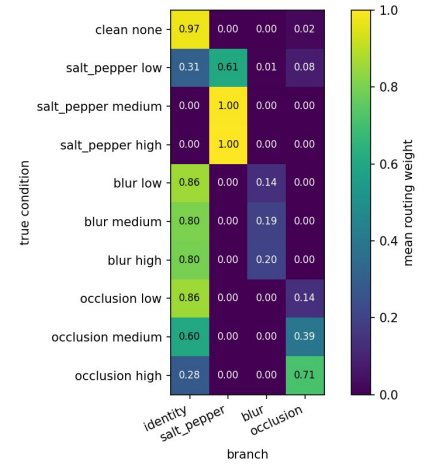


Fig. 12. Mean routing weight per true condition and branch.

TABLE X
SUMMARY OF ALL RESTORATION SYSTEMS ON THE 36,690 TEST ENTRIES (PSNR IN DB / SSIM).

Condition	Input	Task 1	Task 2	Task 3
Clean	– / 1.000	18.82 / 0.497	99.49 / 0.997	51.31 / 0.999
Salt-and-pepper	16.40 / 0.367	18.83 / 0.495	18.58 / 0.501	19.63 / 0.540
Gaussian blur	29.14 / 0.843	18.89 / 0.496	18.61 / 0.501	27.10 / 0.815
Occlusion	13.56 / 0.708	17.92 / 0.471	18.13 / 0.480	17.88 / 0.721
All corrupted	19.70 / 0.639	18.55 / 0.487	18.44 / 0.494	21.54 / 0.692

Style 3 reaches 0.585, but Style 2 reaches only 0.377 and 12.17 dB. Style 2 ground truth sketches have dense dark strokes for hair and contours. A generated stroke that is a few pixels off the true stroke is penalised twice, once where it is and once where it should be, so pixel metrics punish plausible sketches. Style 3 has only 46 test pairs, so its high score is uncertain.

Figure 20 shows that the style condition works. The same photo gives faint thin lines in Style 1, heavy dark hair and contours in Style 2, and an intermediate result in Style 3, which matches the ground truth conventions. Figure 19 shows representative examples and Fig. 21 shows failures. In the failure cases the generated Style 2 sketches have recognisable faces but smoother hair than the ground truth, and the SSIM falls to 0.17 to 0.20. A second failure type appears in Fig. 20, where a head turned to the side produces a distorted face probably because few training faces are strongly turned. The application also reveals an out-of-domain failure. A dog photo produces a faint and meaningless sketch (Fig. 24), because the model has only seen human faces.

IX. OPTUNA SUMMARY

Tables XII and XIII summarise all five studies. The studies are small because of the time budget, with 5 to 20 trials, so they show reasonable settings and not a proven optimum. Pruning removed 6 to 10 trials per study. The records are in the repository as trial tables and SQLite databases.

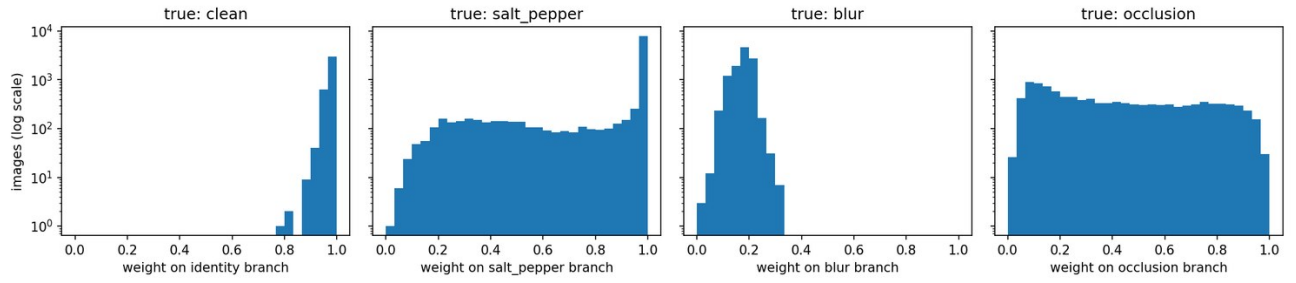


Fig. 13. Distribution of routing weights per branch.



Fig. 14. Cases where the gate spreads weight over several branches.

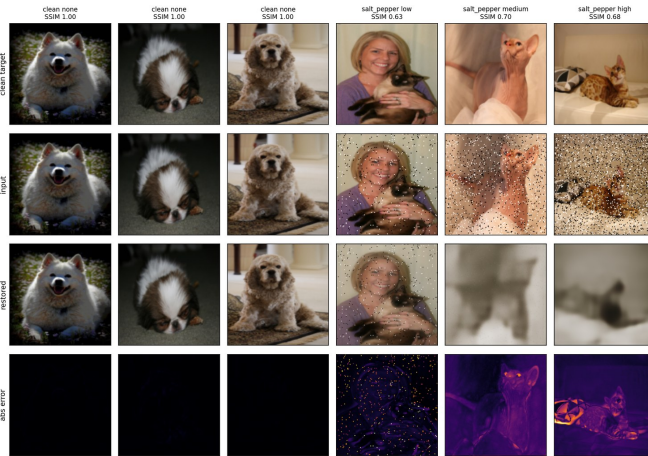


Fig. 15. Task 3 examples.

X. APPLICATION DESIGN AND DEPLOYMENT

A. Design in Google Stitch

The visual design was created in Google Stitch before any frontend code. Figure 22 shows the four Stitch screens. The React and Tailwind application keeps the same page structure, with a sidebar for the four workspaces, a settings card, and image panels.

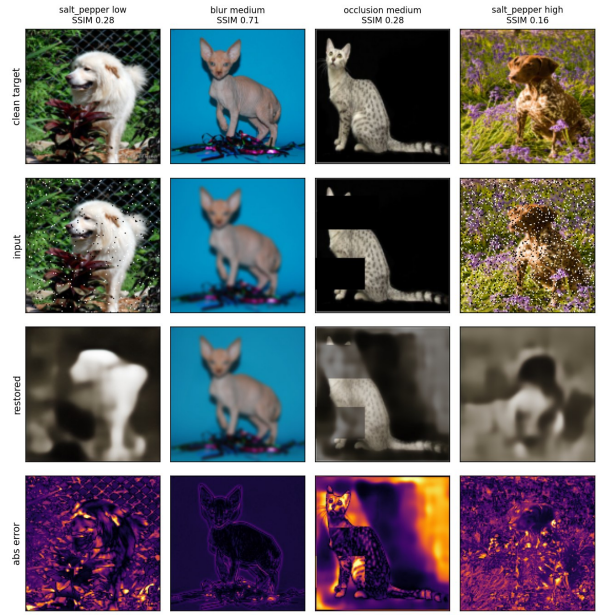


Fig. 16. Task 3 failure cases.

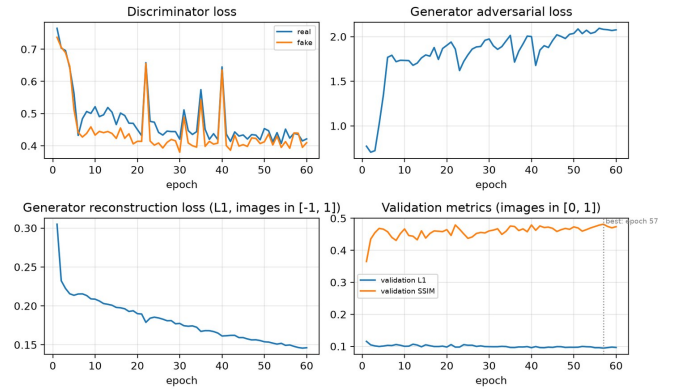


Fig. 17. Task 4 training curves for generator and discriminator.

B. Architecture

The React frontend is built with Vite and served by nginx, which also proxies the API. The FastAPI backend exposes a health check and one endpoint per task: universal restoration,

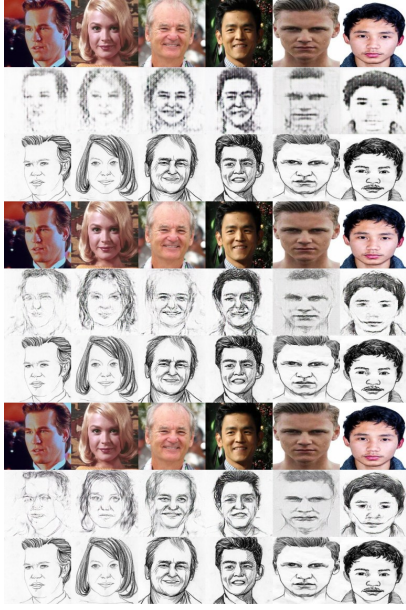


Fig. 18. Fixed validation samples at epochs 10, 20, 30, 40, 50, and 60.

TABLE XI
TASK 4 TEST RESULTS ON 1,045 FS2K TEST PAIRS.

Group	Pairs	L1	SSIM	PSNR
All	1045	0.1095	0.459	15.48
Style 1	618	0.0820	0.500	17.30
Style 2	381	0.1592	0.377	12.17
Style 3	46	0.0670	0.585	18.44

hard routing, soft mixture, and sketch generation. It checks the file type (JPEG, PNG, or WebP) and the size (10 MB), converts to RGB, resizes to 128×128 , applies the requested corruption using the same code as the experiments, runs ONNX Runtime on CPU, and returns images, routing information, and timings. The hard-routed endpoint also returns the four classifier probabilities and the selected expert, and the soft endpoint returns the four routing weights.

All seven models are exported to ONNX and compared with their PyTorch versions by an automatic check script. The models are not stored in Git. They are mounted read-only into the backend container, and a download script fetches them from a documented Google Drive folder. Docker Compose starts the backend, waits for its health check, and then starts the frontend, so the whole application starts with one command. We tested the containerised build. The backend loaded all seven models, the health badge reported them, and requests to the universal, hard, soft, and sketch endpoints returned success. The observed CPU inference time was between 29 ms and 96 ms per restoration request in our screenshots.

C. Workspaces

Each restoration workspace lets the user upload an image, choose a corruption and severity or upload an already corrupted image, and shows the original, the corrupted input,

TABLE XII
OPTUNA STUDIES.

Study	Trials	Completed	Pruned	Best trial	Best value
Task 1	20	14	6	15	0.3102
Task 2 classifier	15	5	10	0	0.9986
Task 2 specialists	5	4	0	0	0.3331
Task 3	8	5	3	0	0.1454
Task 4	12	6	6	0	0.3039

TABLE XIII
SELECTED HYPERPARAMETERS.

Study	Best parameters
Task 1	lr = 0.000263, batch_size = 16, latent_dim = 256, base = 48, dropout = 0.0323, alpha = 0.508
Task 2 classifier	lr = 0.000357, batch_size = 16, channels = small, dropout = 0.433, weight_decay = 0.000254
Task 2 specialists	lr = 0.000357, batch_size = 16, latent_dim = 256, base = 32, alpha = 0.509
Task 3	lr = 3.57e-05, temperature = 0.5, lambda_ce = 0.0205, lambda_bal = 0.00205, alpha = 0.526
Task 4	lr_g = 0.000154, lr_d = 0.000863, batch_size = 8, base = 64, dropout = 0.301, embed_dim = 32, lambda_l1 = 121

the restored output, the settings, the time, and the PSNR of input and output. The sketch workspace accepts an upload or a webcam capture, a style choice, and offers download of the result. Figures 23 and 24 show the running application.

D. Experiment tracking

Every training run and Optuna trial is logged to MLflow with hyperparameters, per-epoch losses, validation metrics, and, for Task 4, sample images. One SQLite database is kept per task in the repository and can be opened with the MLflow user interface. The demonstration video shows these records.

XI. LIMITATIONS

- The autoencoders compress to a small vector with no skips, so reconstructions are soft and desaturated. Tasks 1 and 2 score below the corrupted input on average and are clearly worse on blur and on clean images for Task 1.
- The Task 3 blur expert is effectively inactive, so the mixture does not restore blur at all and loses 2 dB against the input on that condition.
- The Optuna studies are small. The specialist study had one failed trial, and Task 3 had 8 trials. Each result comes from one run with one seed, so no confidence intervals are reported.
- FS2K Style 3 has 46 test pairs and the dataset copy is missing one file. Style 2 scores are dominated by stroke misalignment, so SSIM understates quality.
- The sketch model handles human frontal faces and fails on out-of-domain inputs such as animals and strongly turned heads.
- Inference is only tested on CPU with ONNX Runtime at 128×128 resolution.

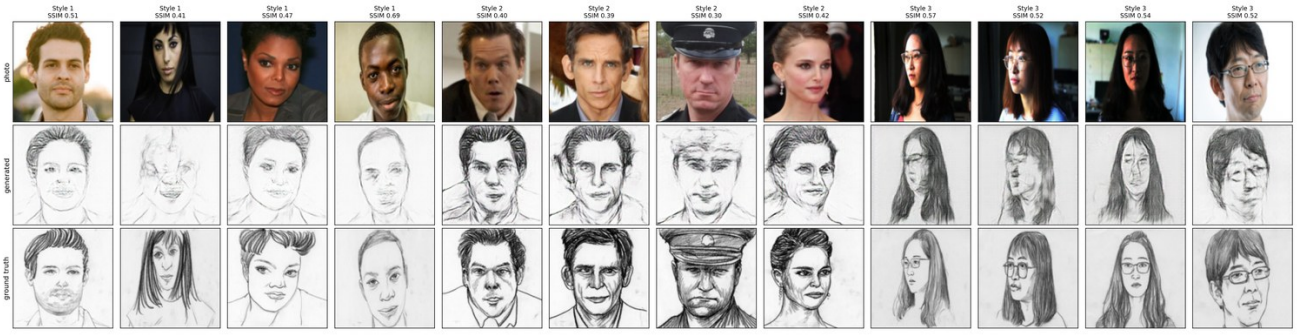


Fig. 19. Task 4 examples: photo, generated sketch, and ground truth.

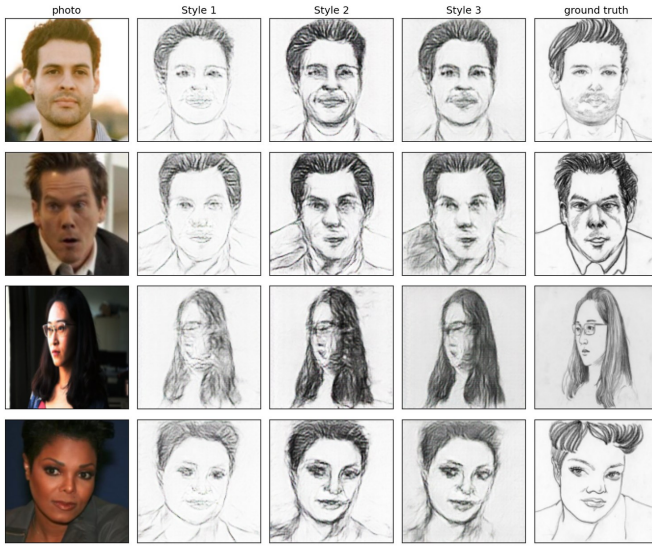


Fig. 20. Style control. The same photo in the three styles next to the ground truth.

Natural next steps are adding limited skip connections or a larger latent to raise the reconstruction ceiling, training the blur expert with a sharpening objective that starts from the input, and running several seeds.

XII. CONCLUSION

Three restoration strategies and a conditional sketch GAN were built, tuned, evaluated on fixed protocols, and deployed in one containerised application. The main result is that the bottleneck sets a reconstruction ceiling of about 18.8 dB that dominates both the universal and hard-routed systems, so perfect routing did not help. Only the soft mixture, which can pass an input through unchanged, beat the do-nothing baseline (21.54 dB against 19.70 dB), and its routing shows that the gate learned to avoid the weak blur expert. The sketch GAN produces plausible and controllable style output with SSIM 0.459, and its weaknesses come from stroke-level metrics and a narrow training domain.

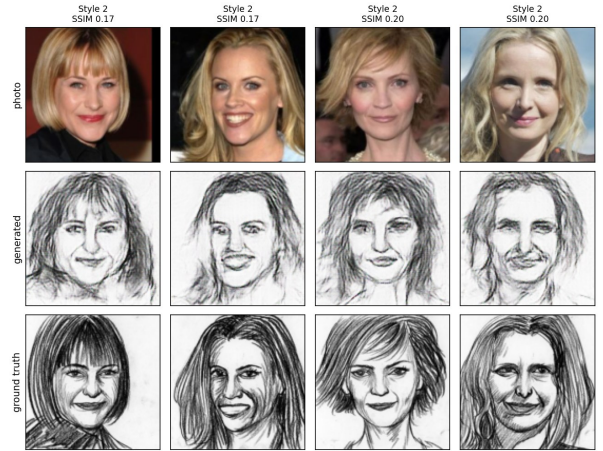


Fig. 21. Task 4 failure cases in Style 2 with low SSIM.

APPENDIX A AI TOOL USE

Claude, an AI assistant from Anthropic, was used throughout the project. It was used to plan the work in stages, draft and debug the Python, FastAPI, React, Docker, and LaTeX code and the README, suggest Optuna search spaces, and draft this report from the saved result files. Google Stitch was used to generate the first interface designs. The effects were faster scaffolding and fewer setup errors. The AI also made mistakes that were found and corrected, including wrong diagnoses of a training problem, code blocks that did not reach the machine, and miscounted figures. The author ran every experiment on Colab and Kaggle, checked the numbers in this report against the result files, tested the application, and is responsible for the submitted content. Training, evaluation, and the results are real runs and not generated text.

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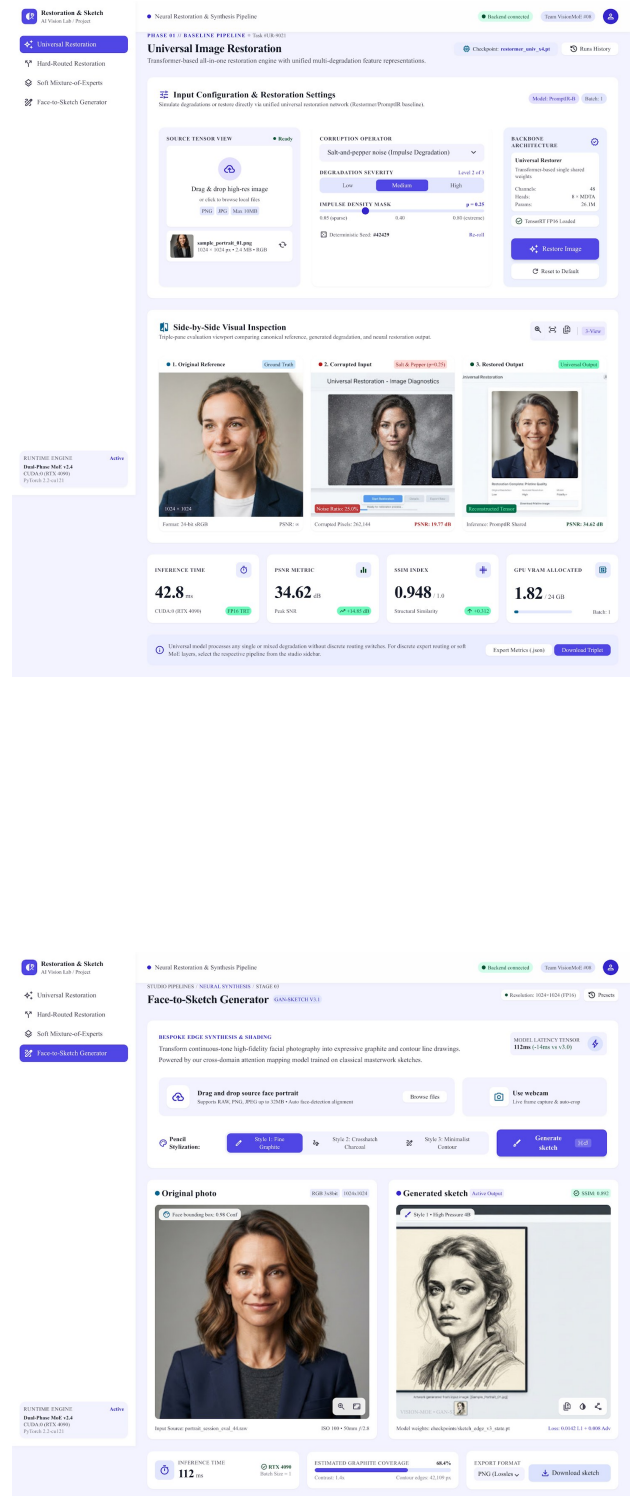
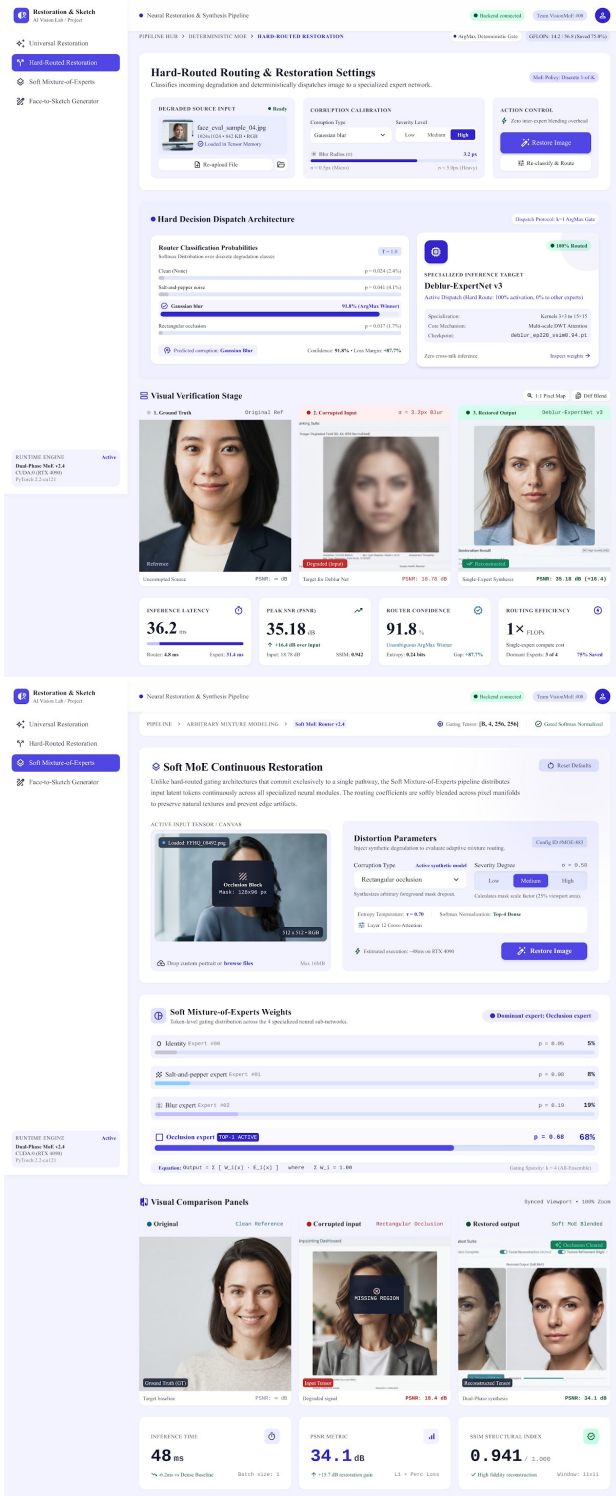


Fig. 22. Original Google Stitch designs for the four workspaces.

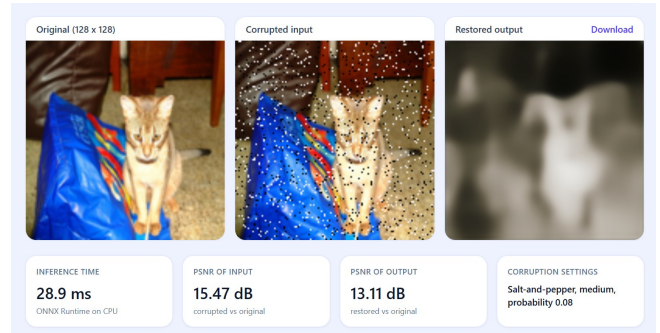
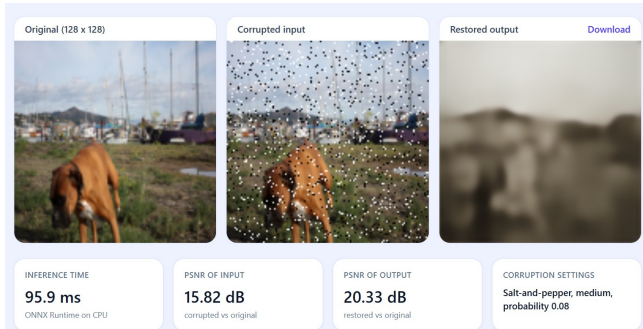


Fig. 23. Restoration result panels for salt-and-pepper noise at medium severity. Left: a successful case (15.82 to 20.33 dB). Right: a failure where the output (13.11 dB) is worse than the corrupted input (15.47 dB).

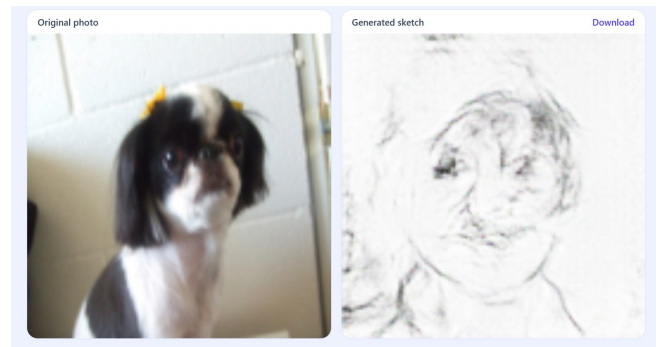
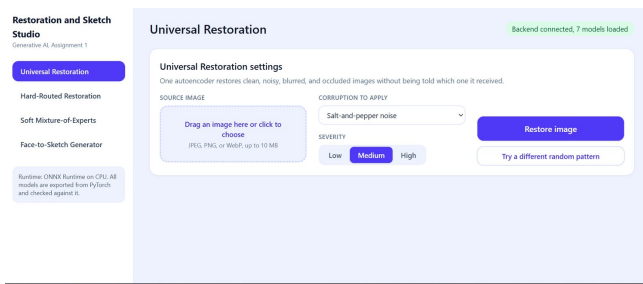


Fig. 24. Left: the Universal Restoration workspace with the backend status badge reporting seven loaded models. Right: the sketch workspace given a dog photo, which is outside the training domain.